Fortress (Military Base / Defensive Warfare & Siege)

Executive Summary

Fortress is a hardened **military-base battlefield** built for **defensive warfare, siege operations, layered fortifications, and autonomous defensive coordination**. The terrain features perimeter walls, watchtowers, bunkers, artillery emplacements, vehicle yards, command centers, kill-zones, and multi-layer defensive rings. The map emphasizes Obsidian Protocol's autonomous defensive logic, sensor-network routing, comms-relay warfare, and logistics under siege conditions. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10x10 m physical area representing ~1 km² of fortified terrain. Key features include layered defenses, breach points, interior strongholds, and siege-driven chokepoints.

Design Goals & Gameplay

Layered Defensive Warfare

Fortress prioritizes multi-ring defense: outer walls, inner bunkers, command citadel. Attackers must breach sequential layers while defenders coordinate autonomous countermeasures.

Siege Operations & Breach Mechanics

Artillery, breaching units, drones, and armored vehicles play major roles. Walls, gates, and bunkers can be destroyed or partially collapsed.

Autonomy & Defensive Coordination

Autonomous units prioritize chokepoints, fallback positions, and counter-battery fire. Units adapt to breached walls, destroyed gates, and shifting defensive priorities.

Sensor & Information Warfare

Smoke, dust, explosions, and bunker occlusion distort sensors. Relay towers on watchtowers restore clarity. Destroying relays creates blind pockets inside the base.

Logistics & Resources

Fuel depots, armories, and vehicle yards serve as resupply nodes. Convoys must navigate interior roads and avoid ambush-prone breach corridors.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between siege units, defensive armor, or autonomous infantry. Blue deploys from the southwest outer gate; Red deploys from the northeast command citadel.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area ≈ 1 km² fortified zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit ≈ 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Outer Gate Command) — (0,0), 150×150 m Gatehouse, barricades, armored staging area.
Red HQ (Command Citadel) — (1000,1000), 150×150 m Hardened command tower with radar mast and AA emplacements.
Central Parade Ground — (500,450), 300×300 m Open interior zone; ideal for armor and artillery duels.
Armory & Vehicle Yard — (250,800), 120×120 m Fuel tanks, repair bays, vehicle shelters.
Bunker Network — (800,300), 200×200 m Interior tunnels, firing ports, hardened defensive positions.
Outer Wall Line — (0,450 → 400,900) Primary defensive ring; breachable gates and towers.
Inner Road Route — (300,0 → 450,350) Safer interior path with partial cover.
Watchtower Relay A — (650,150), +240 m Long-range sensor vantage point.
Watchtower Relay B — (150,650), +260 m Comms relay hub and counter-battery node.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Outer Wall Line

Primary chokepoint; attackers must breach gates or collapse wall segments.

Bunker Network

Interior flanking routes; extremely strong defensive positions.

Parade Ground

Open sightlines; ideal for armor and artillery.

Inner Road Route

Predictable but safe; vulnerable to overwatch from bunkers.

Sightlines

Watchtower Relay A provides long-range sensor coverage (~350 m). Smoke and dust create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW outer gate + siege staging.
Red Deployment: NE command citadel + AA emplacements.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain

Walls, bunkers, towers, parade grounds, vehicle yards, interior roads.

Vegetation

Sparse shrubs, grass patches, military landscaping.

Buildings & Props

Military: AA guns, radar dishes, bunkers, watchtowers. Industrial: Generators, pipelines, fuel tanks. Logistics: Trucks, crates, fuel drums, drones.

Effects

Smoke, sparks, dust, explosions, fire bursts.

Audio

Artillery, machinery, alarms, radio chatter.

LOD & Budget

Asset Cate	Count	Tri/Item	Total Tris
Walls/Tow	3	200k	600k
Bunkers/I	4	80k	320k
Buildings/	15	50k	750k
Roads/Pat	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. Prototype Layout — Month 1 Block-out walls, bunkers, parade ground, AR anchor test.
2. Core Systems — Month 2 NavMesh, defensive autonomy behaviors, siege logic.
3. AR Integration — Month 3 Anchors, plane detection, basic interactivity.
4. Content Fill — Month 4–5 Final models, textures, lighting, audio.
5. Gameplay & Polish — Month 6–7 Objectives, balance, VFX (smoke, sparks, explosions).
6. Testing & QA — Month 8 Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify defensive logic, fallback behavior, breach-zone navigation.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, siege logic, deployment budget enforcement, anchor drift, sensor accuracy.